Tribhuvan University

Faculty of Science and Technology

OFFICE OF THE DEAN

Digital Logic

CSIT — 1st Semester — 2081 — Answer Sheet

Subject Code: CSC116

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Differentiate between synchronous and asynchronous counter. Design a 3-bit synchronous binary counter using T Flip Flop. Draw its timing diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Explain De-Morgan's Law. Simplify the Boolean function F(P,Q,R,S)= Π(0,1,4,5,11,14,15) and d(P,Q,R,S)= Σ(2,3,7,8,9,13) using K-Map in both SOP and POS form.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Explain design procedure of combinational circuits. Design a combinational circuit with three inputs x, y, and z, and three outputs, A, B, and C. When the binary input is 0, 1, 2, or 3, the binary output is one greater than the input. When the binary input is 4, 5, 6, or 7, the binary output is one less than the input.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Given A=46 and B=35 represent them in binary and perform A-B using 1's complement method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is Multiplexer? Design 8 to 1 Multiplexer with low level Multiplexers.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Write about D flip flop with necessary circuit, block diagram, characteristic table and equation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Simplify F(A,B,C,D) = Π(1,3,4,6,9,11,12,14) and realize the equation using NOR gates only.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Mention different types of shift registers. Explain SIPO with timing diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is decoder? Describe the 3 to 8 line decoder circuit.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain state diagram, state table, state reduction and state assignment with suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Design a 2 bit asynchronous binary counter using T Flip Flop. Draw its timing diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes on:a) Encoderb) Error detection codes

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]
